Tesi Doctoral

Lattice and Topology

preparada per

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IFIC (Universitat de València – CSIC)

Departament de Física Teòrica

Programa de Doctorat en Física – 3126

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Directors: Prof. Pilar Hernández Gamazo & Dr. Alberto Ramos Martínez

Ich ging im Walde So für mich hin,
Und nichts zu suchen, Das war mein Sinn.

Im Schatten sah ich Ein Blümchen stehn,
Wie Sterne leuchtend, Wie Äuglein schön.

Ich wollt es brechen, Da sagt es fein:
Soll ich zum Welken Gebrochen sein?

Ich grub's mit allen Den Würzlein aus.
Zum Garten trug ich's Am hübschen Haus.

Und pflanzt es wieder Am stillen Ort;
Nun zweigt es immer Und blüht so fort.

J. W. von Goethe

Alberto Ramos Martínez, investigador distingit (CIDEGENT) de la Universitat de València, i **Pilar Hernández Gamazo**, catedràtica de la Universitat de València, **certifiquen**:

que la present memòria **Lattice and Topology** ha sigut realitzada sota la seua direcció a l'Institut de Física Corpuscular, centre mixt de la Universitat de València i del CSIC, per **David Albandea Jordán** i constitueix la seua Tesi per a optar al grau de Doctor en Física.

I perquè així conste, en compliment de la legislació vigent, presenten en el Departament de Física Teórica de la Universitat de València la referida Tesi Doctoral, i firmen el present certificat.

València, a 28 de Juliol del 2023,

Jacobo López Pavón Nuria Rius Dionis

Preface

List of Publications

This doctoral thesis is based on the following publications:

Notation

The natural unit system $\hbar = c = k_B = 1$, and thus $h = 2\pi$, is used throughout the thesis. This implies the dimensional relation [Energy] = [Mass] = [Temperature] = [Time^{-1}] = [Length^{-1}]. The Fermi coupling constant is $G_F = 1.1663787(6) \times 10^{-5}\,\mathrm{GeV}^{-2}$ [1]. The vacuum expectation value (vev) of the Standard Model Higgs is taken to be $v = 174.104\,\mathrm{GeV}$. The metric with signature $\mathrm{diag}(-1, +1, +1, +1)$ is used. The reduced Planck mass is $M_{\mathrm{Planck}} = \sqrt{\hbar c/(8\pi G_{\mathrm{Newton}})} = \sqrt{1/(8\pi G_{\mathrm{Newton}})} = 2.435 \times 10^{18}\,\mathrm{GeV}$. A variable written in bold format, e.g. $\mathbf{x}$, represents a tuple or vector. Implicit summation over repeated indices is assumed.

Abbreviations

Abbreviation	Meaning
SM	Standard Model of particle physics
BSM	Beyond the Standard Model of particle physics
DM	Dark matter
CMB	Cosmic microwave background
CP	Charge parity
QCD	Quantum chromodynamics
LHC	Large Hadron Collider
C	Charge
LSS	Large scale structure
vev	Vacuum expectation value
QED	Quantum electrodynamics
CERN	Conseil Européen pour la Recherche Nucléaire
SK	Super-Kamiokande
SNO	Sudbury Neutrino Observatory
B	Baryon
L	Lepton
CKM	Cabibbo–Kobayashi–Maskawa
P	Parity
HNL	Heavy neutral lepton
PDG	Particle Data Group
PMNS	Pontecorvo-Maki-Nakagawa-Sakata
NC	Neutral current
CC	Charge current
MSW	Mikheyev-Smirnov-Wolfenstein
SSM	Standard Solar Model
T2K	Tokai to Kamioka
NH	Normal hierarchy
IH	Inverted hierarchy

Abbreviation	**Meaning**
HK	Hyper-Kamiokande
DUNE	Deep Underground Neutrino Experiment
T2HK	Tokai to Hyper-Kamiokande
LNV	Lepton number violation
$0\nu\beta\beta$	Neutrinoless double-beta decay
$2\nu\beta\beta$	Two neutrino double-beta decay
NME	Nuclear matrix element
ΛCDM	Standard model of cosmology
FLRW	Friedmann-Lemaître-Robertson-Walker
BBN	Big Bang Nucleosynthesis
BAO	Baryonic acoustic oscillations
WMAP	Wilkinson Microwave Anisotropy Probe
DE	Dark energy
MES	Minimal extended seesaw
MCMC	Markov chain Monte Carlo
T	Time
CPT	Charge parity time
GUT	Grand Unified Theory
EWPT	Electroweak phase transition
LN	Lepton number
wLNV	Weak lepton number violating
LNC	Lepton number conserving
sLNV	Strong lepton number violating
sHC	Strong helicity conservation
wHC	Weak helicity conservation

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Eh?

N. Bohr

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Bibliography

[1] **Particle Data Group** Collaboration, R. L. Workman and Others, *Review of Particle Physics*, *PTEP* **2022** (2022) 083C01.

Part II

Resum de la tesi

Resum A

Objectius

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